Exercise 1 – IPW From Scratch

Setting and data

Consider a hypothetical observational study investigating the effect of initiating an antihypertensive medication on the risk of experiencing a cardiovascular event within one year.

Patients are assessed at baseline, when their demographic and clinical characteristics are recorded. Based on these characteristics and clinical decisions, some patients initiate the medication, while others do not.

For simplicity, treatment is assigned at baseline and is assumed to remain unchanged throughout the one-year follow-up period.

The dataset contains the following variables:

- X_1: age at baseline (years);
- X_2: baseline systolic blood pressure (mmHg);
- $X_3 \in \{0, 1\}$: diabetes diagnosis at baseline (1 = yes, 0 = no);
- $A \in \{0, 1\}$: initiation of antihypertensive medication (1 = treatment, 0 = no treatment);
- $Y \in \{0, 1\}$: occurrence of a cardiovascular event within one year (1 = yes, 0 = no).

Age, baseline blood pressure, and diabetes status may influence both the decision to initiate treatment and the risk of a cardiovascular event. For simplicity, we assume that these three baseline covariates are mutually independent.

The data are simulated from a known data-generating process, but the analysis should initially be approached as if only the observed variables (X_1, X_2, X_3, A, Y) were available.

The goal of this exercise is to estimate the **average treatment effect (ATE)** using inverse probability weighting (IPW).

Data availability. To complete this exercise, use the dataset `EX1_Data_Observed.RDS`, available in the GitHub repository.

a) Causal structure

Draw a directed acyclic graph (DAG) representing the assumed relationships among X_1, X_2, X_3, A, and Y.

Based on the DAG:

1. Identify the causal effect of interest.
2. Identify the variable(s) that need to be adjusted for to estimate this effect. Identify the confounder(s), if any.
3. Explain why comparing the observed outcome between treated and untreated individuals without adjustment may not estimate the causal effect of treatment.

b) Assumptions

Consider the three main assumptions required to identify the causal effect from the observed data:

- consistency;
- conditional exchangeability;
- positivity.

For each assumption:

1. State the assumption formally using potential-outcome notation where appropriate.
2. Explain its meaning in the context of this specific study.
3. Briefly describe what could constitute a violation of each assumption.

c) Target estimand

Let Y^1 and Y^0 denote the potential outcomes under treatment and no treatment, respectively.
The target estimand is the average treatment effect:

$$\text{ATE} = E\left[Y^1 - Y^0\right] = E[Y^1] - E[Y^0].$$

Answer the following questions:

1. What does the ATE represent in this particular problem?
2. Why can the ATE not generally be calculated directly from observational data?
3. Calculate the crude difference in the observed outcome proportion between treated and untreated individuals, $E[Y \mid A = 1] - E[Y \mid A = 0]$. Is this quantity necessarily equal to the ATE? Explain.

d) Estimating the propensity score

The propensity score is defined as

$$e(\mathbf{X}) = P(A = 1 \mid \mathbf{X}),$$

where $\mathbf{X} = (X_1, X_2, X_3)$.
Using the observed data:

1. Fit a logistic regression model for treatment assignment using the baseline covariates as predictors.
2. Obtain the estimated propensity score $\hat{e}(X_i)$ for every individual.
3. Summarize the distribution of the estimated propensity scores separately for treated and untreated individuals.
4. Produce an appropriate visualization showing the propensity-score distributions in the two treatment groups.
5. Based on the estimated propensity scores, discuss whether there appears to be adequate overlap between the treatment groups and whether positivity problems may be present.

e) Constructing inverse probability weights

For estimation of the ATE, define the inverse probability weight for individual i as

$$w_i = \frac{A_i}{\hat{e}(X_i)} + \frac{1 - A_i}{1 - \hat{e}(X_i)}.$$

Without using a dedicated IPW package:

1. Calculate the weight for every individual.
2. Explain why treated and untreated individuals receive different forms of the weight.
3. Summarize the distribution of the weights.
4. Visualize the weight distribution and identify whether any individuals receive particularly large weights.
5. Explain the relationship between extreme propensity scores and extreme inverse probability weights.

f) Checking covariate balance

An important purpose of weighting is to create a pseudo-population in which treatment assignment is no longer associated with the measured confounders.

Assess the balance of the baseline covariates between treated and untreated individuals:

1. Before applying the IPW weights.
2. After applying the IPW weights.

For each covariate, calculate the standardized mean difference (SMD) before and after weighting. The SMD is defined as

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{(s_1^2 + s_0^2)/2}},$$

where $\bar{X}_a$ and s_a^2 denote the mean and variance of the covariate in treatment group $A = a$, respectively.

For the weighted SMD, replace the group means with their weighted counterparts, while keeping the denominator based on the unweighted variances.

Produce an appropriate visualization comparing covariate balance before and after weighting.

Does weighting improve the balance of the baseline covariates between the treatment groups? Explain why this is important for estimating the causal effect.

g) Estimating the ATE

Using the inverse probability weights:

1. Estimate the mean counterfactual outcome under treatment, $E[Y^1]$.
2. Estimate the mean counterfactual outcome under no treatment, $E[Y^0]$.
3. Use these estimates to obtain the IPW estimate of the ATE.

Finally, compare the crude and IPW estimates with the true sample ATE, calculated using the potential outcomes dataset available in `EX1_Data_Counterfactual.RDS`.

Discuss why these quantities may differ.

h) Reflection

Based on this exercise, briefly explain:

1. what population the IPW pseudo-population is intended to represent;
2. which parts of the analysis rely on statistical modelling and which parts rely on causal assumptions.